

ETHANOL, THE CLEANEST FUEL TO FIRE FUTURE MOBILITY



One of the big areas of focus at the World Economic Forum 2023, at Davos, is moving away from conventional fossil fuel fired vehicles to vehicles that use more sustainable fuel, and the talk about the role ethanol can potentially play in automobile revolution of India. Excerpts from the interview of Mr Vijay Nirani, MD, MRN Group and TruAlt Bioenergy, on the ethanol story, is brought for the readers.

What is the kind of response that you are getting here at Davos to the idea on the future of automobiles which could possibly be fired by ethanol?

We are seeing so much of interest across the forum and where the talk across the city is on energy. So the world is moving towards a transition from regular fossil fuels to cleaner and greener fuels and it is extremely important to have the transition at the earliest, and that is what the world is looking at. We are getting a very warm reception with many top cap companies envisaging keen interest and we are having very successful talks with them. And they are looking at India as their next very big set-up. After China, the next story is India and the warm reception that they are giving us, speaks volumes of their interest.

There is much talk about electric mobility, how it could be the future, and the role that EVs could possibly play. What potential does ethanol have as a cleaner fuel alternative to fossil fuel?

EVs are the largest solutions to the transition, but EVs are not the only one standalone option, therefore, we will need an array, a bouquet of different bio fuels, and ethanol is one of the biggest source of that. Ethanol comes from a plant base i.e., sugarcane or other plants, and 2G ethanol comes from absolute waste. Ethanol or bio CNG, as against EVs has an interesting cyclability about it when it comes to carbon emission. We at TruAlt Bioenergy which is one of the companies under the MRN Group, make ethanol out of absolute waste, thus creating "Wealth from Waste" and which then enables lakhs of consumers and most importantly the rural population consume it as a primary source of fuel and energy. The biggest solution that I see is that every penny that is earned through ethanol sold by companies like us is that, we pass on the benefit to the root of the food chain i.e., farmers. When farmers get money it is utilised back in the local economy. Vijay Nirani quotes "58% of the Indian

population are dependant on agriculture for their livelihood. Out of this 58%, nearly 80% are small farmers who require credit for their cultivation. With nearly 20% of our GDP dependant of agriculture, when the quality of life of 58% of our population increases, their purchasing power, their investing power and their saving power all increase at the same time, elevating not only their lives but more significantly strategically contributing to the growth of almost the entire nation.

Transport Minister Mr Nitin Gadkari has been pushing the idea of ethanol as a cleaner fuel for a long time now. What is the response that you are getting from the grassroots i.e., the farmers?

The government has done a phenomenal job supporting the agri sector. The national bio fuel policy was a game changer and Mr Gadkari has been the biggest voice in this sector. So farmers have gained awareness on this. If I speak of my area in the north of Karnataka and parts of Maharashtra especially, is that now we are collecting the harvested trash, putting it in bio digesters, generating bio gas out of it and then converting it into methane, which is compressed and sold back to the farmers so that he can utilise it as a fuel to run his tractors. The cost per kg of methane is only Rs 54 under the SATAT scheme as against using diesel, which costs over hundred rupees. He is saving 46 to 50 percent cost on his automation which is an absolute game changer.

Do you see ethanol as a standalone energy or largely as a blend in the future?

We are 20 years late in the ethanol story as compared to Brazil and other advanced nations, where the fuel is already used as a standalone fuel. In the fastest cars in the world, the Formula 1 cars, and hyper cars, ethanol is a standalone fuel, which speaks volumes of its efficiency and it will soon become a reality. The challenge is the availability of ethanol

because we do not have ample ethanol to even achieve a 20% blend rate in vehicles that are currently there. We consume about 4000 crore litres of petrol in India and to blend with 20 per cent of ethanol we require 800-900 crores litres of ethanol. Ethanol needs a larger push so that it becomes a reality for our local economy to grow at large.

How do you assess consumer awareness at this moment?

The current Indian consumer is price conscious. Ethanol is priced at Rs 65 per litre when extracted from sugarcane juice, and Rs 60 per litre when extracted from molasses as compared to petrol which is almost Rs 120-130 per litre in many of the Indian cities. My point is that ethanol is half the cost of your fuel. And the cost of per kilometre running of ethanol is almost about a 20-25 percent lesser than petrol because of efficiency in mileage, which though less is more fuel efficient. Mileage output is 70 per cent as compared to petrol, but the cost of ethanol is 50 percent less. So the 30% drop in mileage is compensated by the 50% cut in the price which is a huge gain to our consumers.

What could potentially be done by the central government to speed-up the option of ethanol?

As of now, the right push towards the automotive sector is what is required. When infrastructure starts coming up ethanol will be taken as a serious fuel, which will enhance the sector and production will increase. My key point to the ethanol story is that it's one of the most sustainable fuels available and it will not only save the FOREX of our country, but strengthen it. The fuel is largely benefitting the local economy to unimaginable levels, so the country grows at large. Ethanol is the future to strengthen our country's economy, elevate our rural population, catapult the agrarian economy, make our planet cleaner and make the lives of everyone safer, in the new sustainable world.